

Browser automation stacks for AI agents: Decision Brief

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Executive Summary

Browser automation stacks for AI agents: coverage 1/3 primary questions, 5 recommendation-safe claims, contradiction count 0.

Decision Context

- Goal: Produce an evidence-backed analytical report for a real decision.
- Subject: Browser automation stacks for AI agents
- Decision context: Choose a primary platform for a production pilot in the next two quarters.
- Geography: global
- Time horizon: current

Evaluation Frame

- browser reliability
- agent ergonomics
- security
- debuggability

Key Findings

- Playwright is the control-stack benchmark: deterministic browser automation, strong debugging primitives, and broad engineering trust. [Evidence: 2f1f1f2c-f168-4631-85a8-3225ceb91c97, b796c5cc-3381-492d-ac2f-91c058d38259]
- Teams still need to build orchestration and sandboxing around it for full AI-agent products. [Evidence: 9e39d6ca-006a-4a5f-9125-e1ee4f028c83]
- Browserbase is strongest when the bottleneck is operating browsers at scale rather than controlling them locally. [Evidence: 790ca465-9ec8-4dac-8fbc-c30e1d8cfdeb, 877207da-6d5a-4079-881e-bb91f6e56ab6]
- Stagehand sits in the agent-ergonomics bucket: it tries to make browser actions more natural for LLM-driven systems while still grounding on deterministic execution. [Evidence: ab84d376-b1c8-43cb-9052-a208879ef018]
- The key question is whether that productivity gain justifies another abstraction layer. [Evidence: df1e25d8-8027-4caa-a9a7-e9a548f8530d]
- It reduces infrastructure burden, but it introduces dependency on a hosted browser layer. [Evidence: c7a7bd7b-c3a4-44d2-a470-3eb6b2707fe8]

Comparative Analysis

- Which stacks are best for deterministic control versus agent abstraction?

- Where does hosted infrastructure reduce engineering load?
- What are the core security and observability tradeoffs?

Recommendation and Decision Posture

- Bounded recommendation: evidence is informative but not strong enough for an unqualified winner call.

Gaps & Risks

- Where does hosted infrastructure reduce engineering load?
- What are the core security and observability tradeoffs?
- Prefer pragmatic operating cost and privacy-aware deployment.

Evidence Coverage

- Covered questions: 1/3
- Contradiction count: 0

Sources

- [Playwright documentation](<https://playwright.dev/>)
- [Browserbase overview](<https://www.browserbase.com/>)
- [Stagehand overview](<https://stagehand.dev/>)